When Better Choice Models Do Not Matter

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Abstract

Before making a model more expressive, it is worth asking whether the errors it is meant to fix are the errors the system actually makes. We develop that question into a working method for preference ranking. The idea is to compute, before fitting anything, an *effect ceiling*: the largest improvement any richer model in a candidate family could deliver, given where the data actually live. The ceiling then anchors an *error budget* that assigns the system's observed error to approximation, estimation, and drift. We apply the method to a question the community keeps raising about Chatbot Arena, which ranks language models with Bradley–Terry despite well known objections to its independence assumption. A richer Thurstonian alternative with flexible noise and native tie handling seems like it should help. The ceiling says otherwise: on the ability gaps real matchups occupy, the two model families never disagree by more than about one percentage point of win probability, so almost no improvement was available to find. A pre-registered audit of 1.67 million votes, replicated on a second release, confirms the ceiling everywhere. What actually limits the system is estimation error, cold-start coverage, and nonstationary drift, each of which exceeds the ceiling by one to two orders of magnitude. The general lesson is that expressiveness pays only where approximation error is the binding constraint, and the method tells you in advance whether it is.

1 Introduction

A recurring situation in applied modeling goes like this. A deployed system uses a simple model. The simple model rests on an assumption that theory says is wrong. A richer model exists that fixes the assumption. Someone proposes switching. The proposal sounds like progress, and evaluating it sounds straightforward: fit both models and see which one wins.

This paper argues that the comparison should start somewhere else. Before fitting anything, one can often compute how large the richer model's advantage could possibly be. The two models are functions, the data occupy a known region, and if the functions barely differ on that region, then no amount of fitting will separate them. We call this computation an effect ceiling. When the ceiling comes out far below the smallest difference that would matter in practice, the comparison is settled before it begins, and attention should move to the errors that remain. Assigning the system's observed error to its actual sources, which we call an error budget, is the second half of the method. Together the two computations answer a question that matters well beyond any single comparison: is this system limited by what its model can express, or by what its data can estimate?

We develop the method on a live case. Chatbot Arena ranks language models by fitting Bradley–Terry to millions of pairwise human votes [2, 3]. Bradley–Terry assumes independence of irrelevant alternatives, an axiom human preferences owe nothing to [12], and the platform's comparison pool grows every month while ties, a fifth of quality votes, sit outside the model's outcome space entirely. The classical remedy is the Thurstonian family [17], made exact for arbitrary noise by Cotton's lattice construction [4], with ties arising naturally as dead heats rather than by parametric patch [5]. The hypothesis that a richer, tie-aware link should rank better writes itself, and to our knowledge nobody had tested it at scale.

The ceiling says the hypothesis never had room to be true. Real matchups concentrate near toss-ups, where every sensible link agrees. On the observed distribution of ability gaps, the best possible advantage of any link in this Thurstonian family over the logistic is between 0.23 and 0.31 times our minimum practical difference, a threshold we derive from a ten-Elo-point reading error. At no observed matchup do the two links disagree by more than about one percentage point of win probability. We prove the general version of this argument as Theorem 2: when two model families are uniformly close on the support of the data, no fitting procedure, and in fact no state of the world, can separate them by more than a computable amount.

The audit confirms the ceiling in every direction we test. Across sixteen months of pool growth, refit stability is statistically indistinguishable between the families. Held-out calibration is practically equivalent, with a real but sub-practical lean toward Bradley–Terry. The two tie mechanisms have provably different shapes, yet fit observed ties equally well, because the data are thin exactly where the shapes diverge. Every comparison replicates on an independent 2025 release. Meanwhile the budget shows where the error actually lives: all methods share the same forty-six percent underconfidence on next-month votes, up to three quarters of next-month votes involve a brand-new model that no ability estimator can score, and the tie rate itself drifts by half over the study period. Each of these exceeds the entire link-choice ceiling by one to two orders of magnitude.

Three findings give the audit its texture. First, the families do differ, but structurally rather than statistically: Davidson's tie parameter provably never touches its decisive predictions, while the lattice's single width parameter buys tie mass at the price of decisive accuracy, a cost we measure (Theorem 1, Section 6.4). Second, the one apparent exception we observed, a cold-start episode in which the steeper link seemed to protect against a bad extrapolation, dissolved under our own audit into a numerical artifact; explicit ridge regularization then delivered the genuine protection the artifact had only imitated (Section 6.3.1). Third, one anomaly survives every attribution attempt we make (Section 7).

Our contributions, in the order we would defend them: a methodology, the effect ceiling, that any researcher comparing two model families can compute tomorrow on their own data; a framework, the error budget, that turns "which model wins" into the more useful question "where does the error come from"; and a case study rigorous enough to trust, in which a large modern preference system is shown to be estimation-limited rather than approximation-limited, with every verdict pre-registered, thresholded, gated, and replicated.

2 Related work

The models we compare descend from a century of work on paired comparison. Thurstone modeled choices as comparisons of noisy latent performances [13, 17]; Bradley and Terry gave the logistic special case its modern form [2], which Luce showed is exactly the model implied by independence of irrelevant alternatives [12]. Harville and Plackett extended the family to multi-entry contests and rankings [9, 14], Davidson added an explicit tie outcome [5], and Elo and Glicko turned the ideas into the online rating systems used everywhere from chess to matchmaking [7, 8]. TrueSkill shares the Thurstonian generative story and fits it by approximate Bayesian inference [10]; we cite it for positioning and do not test it, because its different estimation machinery would confound the link-shape question our design isolates. Ranking models without a latent performance scale, such as the Mallows family, sit outside the random-utility tradition and behave delicately as the number of alternatives grows [1, 6], so we stay inside it.

Cotton's lattice construction deserves a careful sentence, because we use its machinery while testing none of its main claims. The construction computes exact win probabilities for contests among any number of entrants, with arbitrary discretized noise and ties as dead-heat mass, and it remains coherent when entrants are added or removed [4]. That multi-entrant coherence is the paper's primary theorem. This paper is not a test of Cotton's primary theorem. Arena votes are pairwise, and on two entrants the lattice reduces to a one-dimensional curve in the ability difference, directly comparable to the logistic. What we evaluate is that curve and its tie mechanism. Testing field coherence itself would require listwise data that no public release contains.

On the platform side, the Arena leaderboard fits Bradley–Terry over deduplicated anonymous battles with ties given half weight in each direction [3]; our pipeline reproduces the published board to 0.18 Elo points of mean absolute error (Section 5). The platform's methodology has since grown

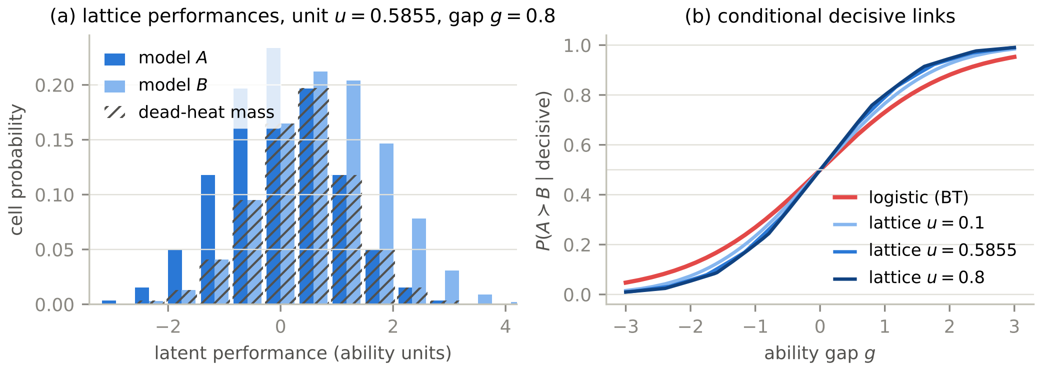

Figure 1: The lattice model of a vote. Panel (a) shows two discretized performance distributions at gap $g = 0.8$ and cell width $u = 0.5855$; a coarse lattice is drawn so the cells are visible, while fits use 500 cells per side. The hatched region is dead-heat mass, the probability that both performances land in the same cell. Panel (b) shows the decisive links this induces at three widths, next to the logistic. The lattice links are steeper at a toss-up, with slopes from 0.290 to 0.337 against the logistic's 0.25, and they steepen as the tie band widens.

style-controlled variants; we analyze the raw preference log and make no claims about style adjustment. A separate line of work documents violations of the axioms these models assume, from positional bias in LLM judging [16] to measured failures of transitivity and independence [18], with social-choice aggregation and listwise objectives proposed in response [11, 15]. The anomaly of Section 7 adds a human-vote observation to that literature: a violation shared by every static model we test rather than distinguishing between them.

3 Two ways to model a vote

Suppose model i has an ability θ_i, and a battle between A and B is decided by which side draws the higher performance $X = \theta + \varepsilon$, with independent noise on each side. Only the difference between abilities matters, so any such model reduces to an increasing curve in the gap $g = \theta_A - \theta_B$:

$$\Pr(A \succ B) = F(g). \tag{1}$$

We call F the link. Bradley–Terry is the logistic link $\sigma(g) = (1 + e^{-g})^{-1}$, and it is the unique link consistent with independence of irrelevant alternatives. Continuous noise leaves no room for ties, and the two families we compare restore them in instructively opposite ways.

The lattice family discretizes the noise onto cells of width u, so that performances take values $x_j = ju$. Writing $a_j(g)$ and b_j for the two sides' cell probabilities, the outcome probabilities are

$$W(g) = \sum_j a_j(g)\, \Pr(X_B < x_j), \qquad D(g) = \sum_j a_j(g)\, b_j, \qquad L(g) = W(-g), \tag{2}$$

and a tie is a dead heat: both draws in one cell, as in Figure 1a. The width u is the family's single tie parameter, and we fit it by profile likelihood. Conditional on a decisive outcome the family predicts with

$$F_u(g) = \frac{W(g)}{W(g) + L(g)}, \tag{3}$$

and two properties of this curve shape everything downstream. It is steeper than the logistic at a toss-up, and it steepens further as u grows, because carving dead-heat mass out of the middle of the distribution sharpens what remains; the logistic's slope is not attainable at any width. It follows that a single parameter controls tie mass and decisive steepness together. The family cannot adjust one without moving the other, a property we will call entanglement and whose price we measure in Section 6.4.

Davidson's family extends Bradley–Terry instead. With tie parameter $\nu \geq 0$ and normalizer $Z_\nu(g) = 2\cosh(g/2) + \nu$,

$$\Pr(A \succ B \mid g) = \frac{e^{g/2}}{Z_\nu(g)}, \qquad \Pr(\text{tie} \mid g) = \frac{\nu}{Z_\nu(g)}, \qquad \Pr(B \succ A \mid g) = \frac{e^{-g/2}}{Z_\nu(g)}. \tag{4}$$

Theorem 1 (Tie orthogonality) *For every $\nu \geq 0$, Davidson's conditional decisive link is exactly logistic:* $\Pr(A \succ B \mid \text{decisive}, g) = \sigma(g)$.

Proof. Conditioning on a decisive outcome cancels the normalizer, leaving $e^{g/2}/(e^{g/2} + e^{-g/2}) = \sigma(g)$. □

The theorem makes the two families a matched pair in the cleanest possible sense. Each has one ability per model and one tie parameter. They differ in exactly one structural respect: Davidson's tie parameter never touches its decisive predictions, and the lattice's always does. Every tie-related result in this paper traces back to this contrast.

A few conventions used throughout. All fits are per-vote maximum likelihood with analytic gradients, and only the link differs between methods. Where an experiment mirrors the production leaderboard, decisive votes count twice in their direction and each tie counts once in each direction through the decisive link; where tie prediction is itself the question, we use the full three-outcome likelihood. Every fit anchors the same reference model, a dated checkpoint present with high volume in every training window, and ability magnitudes compared across links are rescaled so that one reported unit buys the same win-probability change at a toss-up in every method, the same role the familiar 400 over log 10 constant plays for Elo.

4 Effect ceilings and the error budget

The question "is the richer model better" has a prerequisite that is almost never computed: how much better could it possibly be? Both models will be fit to the same votes and scored on the same outcomes. If their links barely differ on the gaps the data contain, the comparison is over before it starts. This section turns that observation into a theorem, a number, and a framework.

Start with the honest version of the best case. Suppose the world really is the richer family: outcomes are generated by some lattice link F_u. The simpler family still gets to fit itself, so the richer family's true advantage is only what the best logistic concedes. With $\hat{P}$ the empirical distribution of fitted gaps, that concession is

$$C \;=\; \min_{s>0} \; \mathbb{E}_{g\sim\hat{P}} \, \mathrm{KL}\big(\mathrm{Bern}(F_u(g)) \,\big\|\, \mathrm{Bern}(\sigma(g/s))\big)\,, \tag{5}$$

and we call C the effect ceiling. The following theorem says the ceiling means what it appears to mean, and that a cruder version of it holds with no assumption about the world at all.

Theorem 2 (Approximation-limited regime) *Let p and p' be two prediction rules mapping gaps to win probabilities, and let $m(g)$ denote the smallest of $p(g)$, $1 - p(g)$, $p'(g)$, $1 - p'(g)$. (i) For every sequence of outcomes whatsoever, the difference in average log loss between predicting with p and with p' is at most $\mathbb{E}_{g\sim\hat{P}}[\,|p(g) - p'(g)|/m(g)\,]$. (ii) If outcomes are generated with true win probabilities $p(g)$, the expected excess log loss of predicting with p' instead of p equals $\mathbb{E}_{g\sim\hat{P}}\,\mathrm{KL}(\mathrm{Bern}(p) \,\|\, \mathrm{Bern}(p'))$. Consequently the expected advantage of the true family over the best member of a competing family equals the effect ceiling of Eq. (5), and predicting with any estimated member of the true family rather than the truth itself only reduces that advantage.*

Proof. For (i), fix one vote with gap g and outcome y. The prediction enters the loss as $\log p$ or $\log(1-p)$, and by the mean value theorem the two rules' losses differ by at most $|p - p'|$ divided by the smaller of the two arguments, which $m(g)$ bounds. Averaging over votes gives the claim. For (ii),

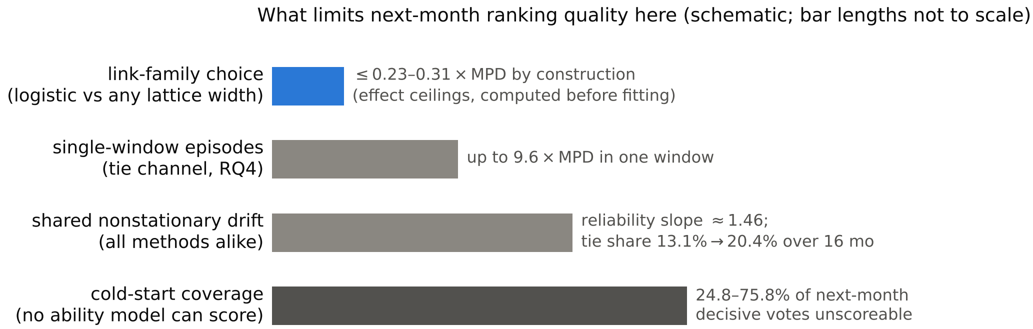

Figure 2: The error budget of the system we study, drawn schematically. The quantity that model choice controls is bounded by construction at a fraction of the practical threshold. The quantities that actually limit ranking quality live on other axes and exceed it by one to two orders of magnitude. Annotated values are measured; bar lengths are not to scale.

take expectations of the log loss under $\mathrm{Bern}(p(g))$: the difference between using p' and using p is exactly the Kullback–Leibler divergence at each gap, and minimizing its average over the competing family gives the ceiling. Estimation error on the true side adds $\mathbb{E}\,\mathrm{KL}(\mathrm{Bern}(p)\|\mathrm{Bern}(\hat{p})) \geq 0$ to the true side's loss. □

Now put numbers in. Real Arena matchups concentrate near toss-ups; the median fitted gap magnitude is 0.31, and every increasing link agrees at a toss-up by construction. On the observed gap distribution, the lattice link at its primary width and its best logistic competitor never differ by more than 0.011 in win probability, and on the average vote they differ by 0.0009. Part (i) of the theorem, which assumes nothing about how outcomes arise, already confines any conceivable log-loss separation to 6.6 times the minimum practical difference defined in Section 5. Part (ii), the sharp in-family computation of Eq. (5), closes it to 0.009 at the primary width, and to at most 0.23 across every width and skew we consider plausible. The tie-mechanism analogue, computed on three-outcome predictions, is at most 0.31 in both truth directions. These are upper bounds on what fitting could ever recover, computed before any experiment ran, and they set the honest expectation for everything in Section 6: practical equivalence, everywhere.

The ceiling is one line of a larger account. A ranking system's error has several possible sources: the model family may be unable to express the truth, which is approximation error; the data may be too thin or too new to pin down the parameters, which is estimation error; and the world may move between training and use, which is drift. Each source has its own instrument. The ceiling bounds the approximation line. Standard errors, coverage counts, and regularization experiments measure the estimation line. Recalibration slopes on held-out predictions measure drift. Filling in the lines for a given system is what we mean by an error budget, and Figure 2 shows the completed budget for Arena: the approximation line is bounded at a fraction of one practical unit, and every other line dwarfs it. The rest of the paper is the audit that fills in that figure, and we would argue the budget, not any single verdict, is the real product of the exercise. A reader comparing logistic regression to a neural network, or Plackett–Luce to Mallows, or Gaussian to Student-t noise, can compute the same two objects on their own data tomorrow, and will know before fitting whether the comparison can matter.

5 The case study: data and protocol

Our main data is the largest public Arena battle log: 1,799,991 anonymous pairwise battles across 129 models, timestamped to the second, from April 2023 through August 2024. The platform's own deduplication retains 1,670,250 battles, and all fits use that subset. Outcomes are a win for either side, a tie, or a tie marked "both bad." Quality ties are 20.45 percent of the votes that are not both-bad, up from 13.1 percent in the first three months, a drift that will matter repeatedly. The replication set is the 2025 release: 135,634 battles across 53 models over fourteen weeks.

Before comparing anything we verified that the pipeline reproduces production. Refitting Bradley–Terry at the published leaderboard's own timestamp reproduces the published ratings to a mean absolute error of 0.18 Elo points with Spearman correlation 0.99997 across all 128 published models. Extending the check to ten published snapshots spanning nine months, the era-appropriate variant achieves between 0.18 and 1.01 Elo points on nine of ten; the tenth remains an unexplained outlier at 2.26, which we report rather than drop.

Verdicts need a threshold, because at a million votes statistical significance is free. We derive a minimum practical difference from a deployment-meaningful error. Leaderboard consumers read ratings at roughly ten-Elo-point granularity; a uniform ten-point error in every matchup translates to an expected excess log loss of 4×10^{-4} nats per vote at a toss-up, and that is our threshold for decisive calibration. The analogous computation for tie prediction, a one-point error in tie probability at the observed twenty percent tie rate, gives 3×10^{-4}. Differences below threshold are practically equivalent no matter how significant.

The protocol then fixes everything that could be fixed in advance. The full pipeline had to pass synthetic gates before touching real outcomes: it detected true effects down to a third of threshold with correct sign, called a positive only when one existed, recovered tie parameters within ten percent, and made no false calls in any world whose true effect was below threshold. One gate result matters later: in a high-noise synthetic world, a correctly specified steep link loses held-out to a misspecified logistic, because point-estimate predictions from steep links are overconfident under parameter noise; calibrated standard errors showed this reversal is inactive at our data's pooled noise levels. Real-data verdicts come from a fixed classifier over thirteen rolling monthly windows, training cumulatively and testing on the following month. A positive verdict requires the pooled effect to clear threshold, ten of thirteen windows to agree in sign, and a window-cluster bootstrap interval, whose effective sample size is thirteen and is always stated, to exclude zero. Equivalence requires the interval to sit inside the threshold band, and significant leans below threshold are reported as leans, never as wins. Anything else is inconclusive and says so. Post-hoc analyses are labeled and never override a pre-registered verdict. Machinery ablations confirm the apparatus is not doing the work: pooled intervals move by under 0.004 threshold units across bootstrap sizes from one thousand to fifty thousand, link curves are invariant to lattice resolution at machine precision, and the profiled tie width moves by under one percent across profile grids. Finally, models absent from training are unscoreable and counted: they are 24.8 to 75.8 percent of next-month decisive votes, a number that belongs in the budget rather than a footnote.

6 What the audit found

6.1 Rankings do not move differently

The first worry about a growing pool is churn: new models enter, the leaderboard refits, and incumbents shuffle. If the logistic link were straining against reality, one might expect its refits to shuffle incumbents more than a flexible link's. So for each adjacent pair of monthly checkpoints we fit both methods on identical cumulative data and compare how each ranks the incumbents, defined as models with at least a thousand votes; an all-models variant behaves identically.

Nothing separates them. The median paired difference in Kendall's τ_b is +0.00026; the lattice is more stable in seven windows, less in four, tied in two; the means are 0.98469 and 0.98358; the largest single-window difference is 0.00713. The picture repeats at all three lattice widths, at one-month and three-month horizons, and for both incumbent definitions. No incumbent moved more than five positions in any window under either method, and ability drift per month is 1.70 to 1.77 Elo-equivalent points for every method. The informative contrast in Figure 3 is vertical, not between the curves: stability itself climbs from τ_b of 0.968 to 0.997 as the pool matures, a swing two orders

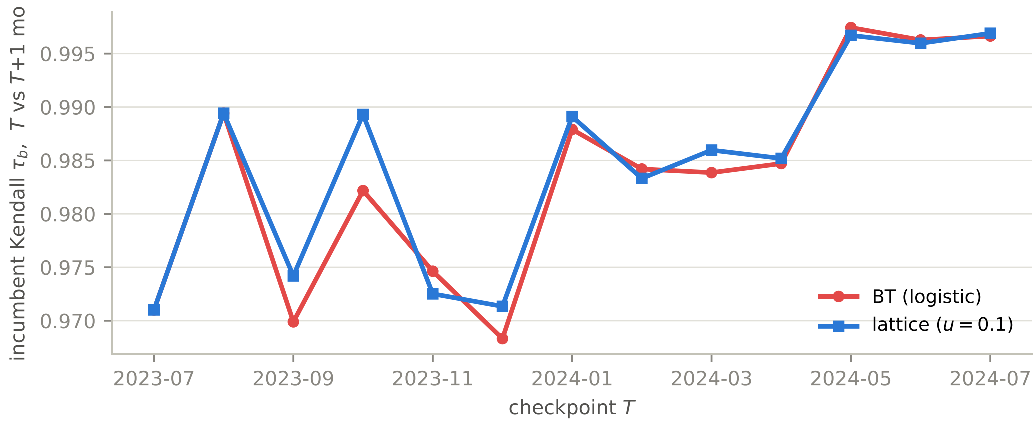

Figure 3: Month-over-month stability of incumbent rankings, measured by Kendall's τ_b, for the two methods across thirteen refit windows. The curves are statistically indistinguishable, and both improve together as the pool matures. Source: `rq1_metrics.csv`.

of magnitude larger than any between-method difference. Stability is governed by sample size, not by link. That is the budget's first entry, and it is an estimation entry.

The ceiling predicted this: where the data live, the links barely differ. That prediction can be checked directly, without fitting any abilities at all.

6.2 The ceiling, confirmed without fitting

Any gap link makes a prediction that overdetermines the data. If the link is F, then for any three models the inverted head-to-head rates must add up: $F^{-1}(p_{AC})$ should equal $F^{-1}(p_{AB})+F^{-1}(p_{BC})$. How badly a triple violates this, scaled by its sampling noise, measures shape misfit in a way that treats every link fairly. We compute the standardized residual

$$z = \frac{F^{-1}(\hat{p}_{AC}) - F^{-1}(\hat{p}_{AB}) - F^{-1}(\hat{p}_{BC})}{\sqrt{v_{AB} + v_{BC} + v_{AC}}}, \qquad v_{XY} = \frac{\hat{p}(1-\hat{p})}{n_{XY}\,[F'(F^{-1}(\hat{p}))]^2}, \tag{6}$$

for each of 786 dense stable triples, meaning all three pairings have at least a hundred decisive votes inside a sixty-day block and all three models are present throughout it. If the link's shape is right and preferences hold still, z is standard normal and the mean of z^2 is one.

The pooled means are 1.202 for the logistic and 1.231, 1.252, and 1.279 for the lattice widths. Synthetic calibration passed, with true links scoring 1.02 to 1.03, and the same synthetic runs measured the test's resolution: worlds generated by either family separate the four links by only 0.01 to 0.04, which is the same order as the real spread. So the test says what the ceiling said, through an entirely different instrument, one that involves no held-out scoring, no point estimates, and no time splits. At the gaps real matchups occupy, no member of this family is distinguishable from the logistic. Two independent derivations of the same bound are better than one.

Shape, then, is settled in-sample. Calibration on future votes could still differ, because estimation noise interacts with steepness even when shapes agree. That is the next entry.

6.3 Calibration: equivalent, and a cautionary tale

We fit both methods per window in the production tie treatment and score log loss on the following month's decisive votes, on the shared set both can score. The sign convention makes positive favor the lattice.

The verdict is equivalence with a lean. Pooled, the lattice loses by 0.121 threshold units at the primary width, by 0.148 and 0.226 at the others, with intervals inside the band and excluding zero,

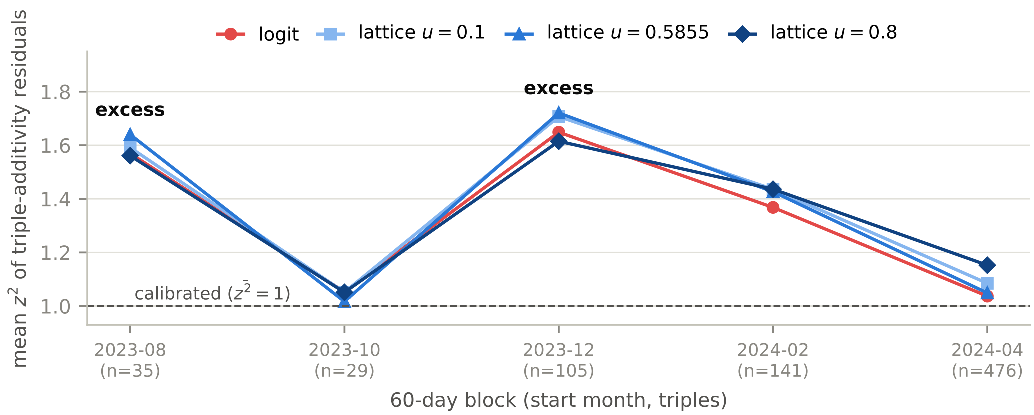

Figure 4: Mean squared additivity residual per sixty-day block, per link. The four links move together, inside the test's measured resolution, while two blocks show a shared excess that no link explains and that Section 7 takes up. Source: `rq2b_blocks.csv`.

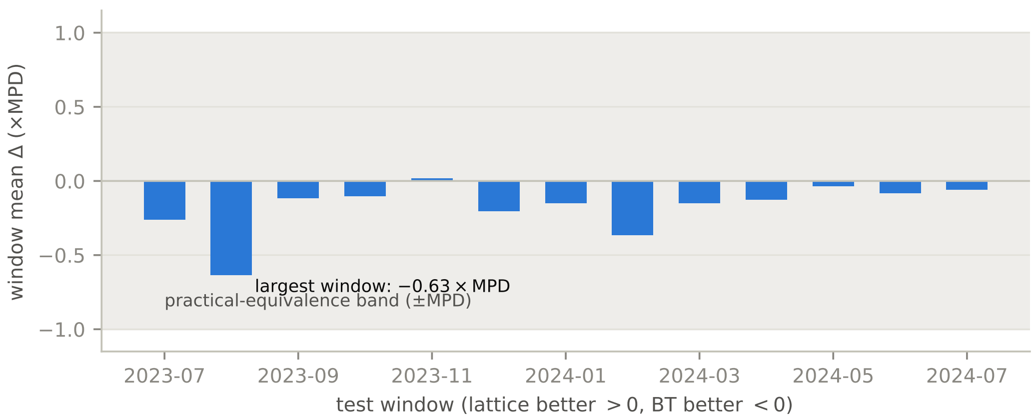

Figure 5: Per-window calibration difference at the primary width, in threshold units. All thirteen windows lie inside the practical-equivalence band. Source: `rq3_window_table_u0.5855.csv`.

and with Bradley–Terry ahead in twelve or thirteen of thirteen windows. Every window sits inside the band; the largest is 0.63 (Figure 5). The lean is real, it is below deployment relevance, and per protocol we report it as a lean. It is largest at the steepest width but not monotone across the shallower two, which is directional and only directional support for the overconfidence mechanism from Section 5. The pre-registered strata agree with the whole: recent entrants show the same sub-practical lean, and in the one stratum where the pre-analysis said a lattice-truth effect could clear threshold, the prediction of +1.58 is excluded decisively, with the measurement at −0.275 and an interval from −1.169 to +0.040.

Two shared measurements swamp all of the above, and they are the budget's drift and coverage lines. Every method's held-out predictions are underconfident by the same factor: recalibration slopes are 1.46 for all four models, and integrating parameter uncertainty into the predictions barely moves anything, shifting the pooled difference from 0.121 to 0.117 and the slopes to 1.46. The miscalibration is the world moving, not the estimator misfiring. And up to three quarters of next-month decisive votes involve a model with no training votes at all, which no link, prior, or smoothing scheme can score.

6.3.1 The episode that wasn't

Our first run of this experiment told a more exciting story, and the story was false in an instructive way. The original verdict was inconclusive: twelve quiet windows and one loud one, in which a model that had entered with two training votes was extrapolated to mid-table by Bradley–Terry while the steeper lattice held it near zero, and Bradley–Terry paid roughly 68 nats when the model was mass-sampled the next month and performed poorly. We read this as implicit shrinkage, a steep link quietly regularizing cold starts, and wrote it up as a mechanism.

The mechanism did not survive its own follow-up. Asked whether explicit ridge regularization could reproduce the protection, we ran a ridge sweep whose near-zero-penalty baseline refused to match our own stored table. The audit that followed located the cause in our earlier lattice implementation, which paired piecewise-linear log-likelihood values with smoothed gradients; the inconsistency, first noticed only as an optimizer nuisance, had been clamping near-zero-data models in the far tails of the curve. With consistent interpolation the episode vanishes. The old code reproduces the loud window exactly, at +5.792 threshold units; the corrected code puts it at +0.017; everything else is byte-identical. The corrected lattice places the cold-start model at +0.29, essentially where Bradley–Terry puts it. All synthetic gates were re-certified under the corrected code, and the equivalence verdict above is what remains.

The retraction sharpened the finding. Steepness is not implicit regularization. But explicit regularization does exactly what the false story promised: ridge on Bradley–Terry shrinks the cold-start model's estimate from +0.38 to zero as the penalty grows from one to three, improves that window by about 5.5 threshold units, and moves established models by at most 0.015. Cold-start protection is real, cheap, and belongs to the estimation line of the budget, not to the model family. We kept the episode in the paper because it demonstrates the method: a pre-registered baseline and one skeptical follow-up caught an artifact that a single headline run would have published.

Calibration cannot separate the families either. By Theorem 1, ties are the one channel where they differ by construction, so ties get the last word.

6.4 Ties: different shapes, identical fit, and the price of entanglement

Both models now fit the full three-outcome likelihood per window, abilities plus one profiled tie parameter each, and are scored on the following month's wins, losses, and ties. Dead heats use the platform's quality-tie label; the both-bad label marks responses below an absolute bar rather than close draws, and appears only in the stress test below. Positive again favors the lattice.

The pooled verdict is inconclusive, and this time the episode is real data. The estimate is +0.838 threshold units with an interval from +0.017 to +2.542, the lattice ahead in ten of thirteen windows, and one window at +9.55 carrying the total. That window decomposes cleanly: its tie votes contribute +407 nats to the lattice against −301 on decisive votes, led by near-peer pairs from the same families, the kind of sibling matchups where ties are common and the lattice's tie curve briefly paid off. A cold-start filter does not touch it, so this is one month of unusual voting rather than an estimation pathology; it immediately precedes one of the two anomaly eras described next, a coincidence we note and do not press.

The structural result is sharper than the verdict. The two fitted mechanisms imply nearly identical tie probability at a toss-up in every window, within about 0.01 of each other, yet their bands have genuinely different shapes: the tie probability halves within 1.67 to 1.68 ability units under the lattice's overlap decay, against 2.84 to 2.93 under Davidson's slow decay, a stable factor of 1.7. Held-out fit cannot tell them apart, with binned errors of 0.0225 and 0.0237, because votes concentrate where the shapes agree (Figure 6). The two tie parameters are not relabelings of one quantity. They carry different shape content that this dataset simply cannot adjudicate, which is a more precise statement than calling them equivalent.

Both tie parameters also drift, and the drift cross-validates. Over the sixteen months the profiled Davidson parameter rises from 0.376 to 0.559 and the lattice width from 0.594 to 0.803, tracking the raw tie share as it climbs from 13.1 to 20.45 percent (Figure 7). Any single pooled tie parameter would average a moving target, which is why every experiment profiles per window. As an internal

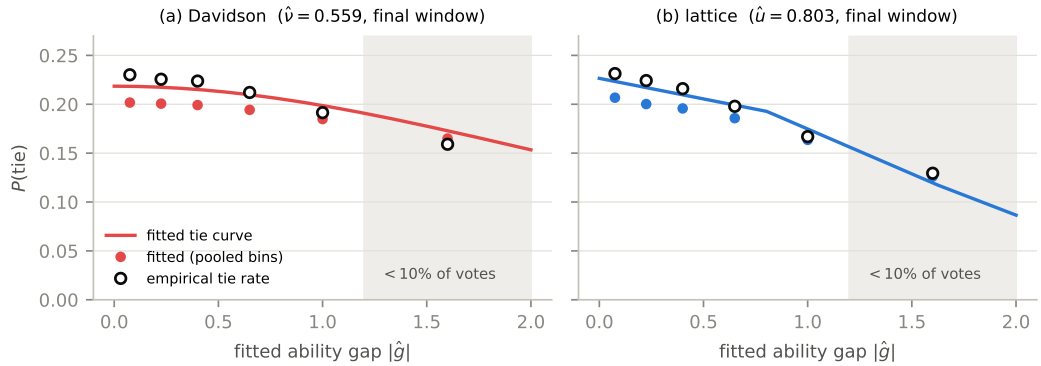

Figure 6: Fitted tie curves against empirical tie rates, pooled with vote weights, each model binned by its own fitted gaps. The mechanisms agree at the origin and decay at structurally different rates, and the region where they separate, shaded, carries under ten percent of votes. Source: `rq4_tie_curve_bins.csv`.

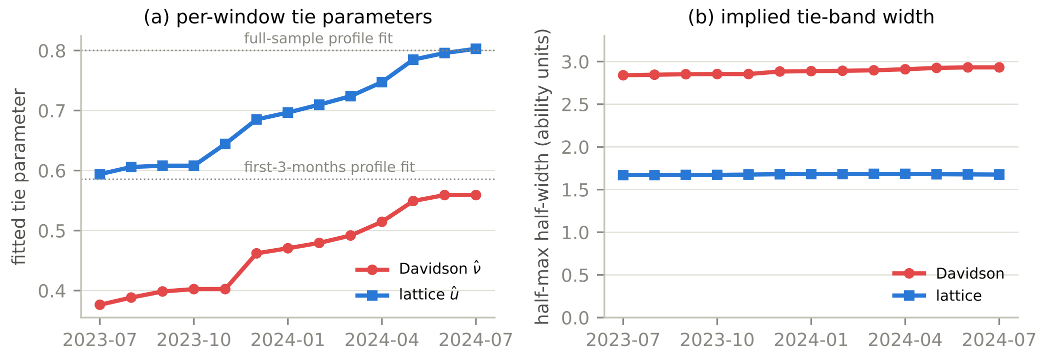

Figure 7: Tie parameters drift together. Panel (a): per-window profiled parameters rise as the platform's tie share rises, and the endpoints reproduce two profile fits computed independently months earlier, shown dotted. Panel (b): the shape difference between the two bands is stable throughout. Source: `rq4_param_trajectories.csv`.

check, the windowed trajectory's endpoints reproduce, to about 0.01, two estimates computed months earlier by a different procedure on the early and full samples.

Finally, entanglement has a measurable price. Theorem 1 says Davidson's tie parameter cannot touch its decisive predictions, and the lattice's must. Forcing both models to absorb both-bad-scale tie mass, about 35 percent of votes, makes the difference concrete. The lattice width pins at its profile boundary in eleven of thirteen windows, not for lack of tie capacity but because its narrow band must over-widen to reach large-gap ties, and on identical held-out decisive votes its log loss degrades by 4.75 threshold units against 2.50 for the Davidson control, which isolates the shared effect of refitting under a changed outcome mix because its decisive link provably cannot move. The excess, roughly 2.2 units, is the measured cost of buying tie mass with decisive accuracy. The measurement is modest and heterogeneous across windows, and the profile boundary was never stress-tested at this tie scale, so we hold it lightly; the conceptual point is Theorem 1's, and the measurement illustrates it.

6.5 The same audit on a second dataset

Every design above transports verbatim to the 2025 release, with twelve weekly windows standing in for thirteen monthly ones; the transport itself was not pre-registered and is labeled accordingly. The full-population fits agree at Spearman 0.999839 with a maximum rank move of one. The calibration comparison again returns equivalence with a sub-practical lean toward Bradley–Terry at all three widths, at 0.084, 0.175, and 0.176 threshold units, with Bradley–Terry ahead in eleven of twelve windows. The tie comparison is again a lattice-leaning inconclusive, at +0.779 with an interval from +0.382 to +1.093. And the profiled width barely moves across fourteen weeks, from 0.689 to 0.693, which confirms the tie drift as a long-horizon phenomenon rather than a constant churn. Different era, substantially different model pool, same nulls, same lean directions, same structure.

7 The anomaly

One signal in this data survives every attribution we attempt, and it deserves its own section precisely because no model we test explains it. In two specific eras, every link fails triple additivity the same way. The blocks beginning in late August and late December of 2023 show mean squared residuals of 1.57 and 1.65, with 13 to 14 percent of triples beyond two standard units against a nominal 4.6, while other blocks sit near one. Splitting blocks in half localizes the first episode to roughly late September and October and shows the second persisting undiminished in both halves, which rules out gradual preference drift within the window. We then checked the platform's deduplication transition, which overlaps only the cleanest block; entry activity, which is anti-correlated with the excess, since the loud blocks have fewer entrants and lower entrant vote share than the quiet ones; language mix, judge counts, votes per judge, tie share, and duplicated-prompt share, none of which separates loud from quiet; and concentration in particular models, which is absent. Because the excess is identical across link shapes, it is a property of the votes, not of any model.

Two explanations remain live. The judge population may shift in composition, since a mixture of individually consistent judges violates additivity in aggregate. Or context may genuinely act on votes, with judgments depending on what else occupies the pool, which is the classical independence violation appearing as a shared failure of every model that assumes it away. Separating the two requires per-judge structure or listwise comparisons, neither of which public releases provide. We had pre-registered an event-study design for exactly this question and descoped it on evidence, since an instrument built on entry proximity is the wrong tool for a phenomenon anti-correlated with entries. We leave the anomaly as an open question, and as a concrete request to preference platforms: judge panel structure and occasional listwise comparisons would let someone answer it, and would also make the multi-entrant theory we deliberately did not test testable at last.

8 Reading the budget

The budget in Figure 2 is the paper's summary. The line that model choice controls was bounded by construction at a fraction of one practical unit, and sixteen months of audit plus a replication never found anything above it. The lines that dominate belong to estimation and drift: a shared factor of 1.46 underconfidence on next-month votes, up to three quarters of votes unscoreable for want of any training data, single months in which one episode moves a verdict by ten threshold units, and a tie rate that drifts by half across the study period. A platform operator reading this budget knows the order of work. Get votes onto new models faster, and shrink their early estimates explicitly, since ridge achieves what no link does. Recalibrate against recent outcomes, since the miscalibration is drift and parameter uncertainty corrections do not touch it. If ties matter to the product, prefer the mechanism whose tie parameter provably leaves decisive predictions alone. And only after all of that, think about the link.

The generalization is the reason we wrote the paper this way. Reward models for RLHF are Bradley–Terry by construction, and proposals to enrich the preference objective, listwise or otherwise, face the same arithmetic whenever comparisons concentrate near toss-ups and the candidate pool churns [11]. Recommendation and matchmaking systems consuming implicit pairwise feedback, sports and game rating systems, and econometric choice models fit to large observational panels all sit in the same position: a simple link under theoretical suspicion, a richer family available, and a comparison that can be priced before it is run. The principle the budget expresses is old statistics wearing modern clothes. Expressiveness buys approximation accuracy, and buying it is only rational when approximation error is the binding constraint. In large, noisy, evolving preference systems, it is not, and one can know that in advance.

We would also defend the audit's conduct as a contribution. The ceilings set expectations before any fit. The synthetic gates certified that the machinery could detect what it claimed to detect. The fixed classifier refused two convenient readings. And the protocol caught our own false mechanism when one skeptical baseline declined to reproduce a stored number, which is exactly the failure mode that makes exciting stories survive into print. Negative results earn their keep when the method that produced them can be reused, and both halves of ours can.

9 Limitations

The disclaimers that matter most are scope. Nothing here tests the multi-entrant coherence property that is Cotton's primary contribution, because pairwise data cannot exercise it. Our conclusions concern two eras of one platform and the specific families compared; TrueSkill-style approximate inference was positioned but not run. The ridge demonstration rests on one natural experiment. The evidence for the overconfidence mechanism is directional rather than monotone. The entanglement measurement leans on a profile boundary that was never stress-tested at both-bad scale. One validation snapshot remains unexplained. And the anomaly is characterized, not solved. Resolving it, like testing coherence, waits on data that platforms do not currently release.

10 Conclusion

We set out to referee a model comparison and ended up pricing it. The price, computed before fitting, was a ceiling of a fraction of one practical unit, and the audit spent 1.8 million votes, a replication, four pre-registered comparisons, one retracted mechanism, and one open anomaly confirming that the price was right. The comparison itself is settled: for ranking language models from pairwise human votes, a richer Thurstonian link buys nothing that matters, and the system's real errors come from estimation, coverage, and drift. The method is not settled, and that is the point. Any proposed jump in model expressiveness can be priced the same way, with an effect ceiling on the family difference and an error budget for everything else, and the jump is worth making only when the ceiling clears the errors that remain. Spend expressiveness where the errors are.

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